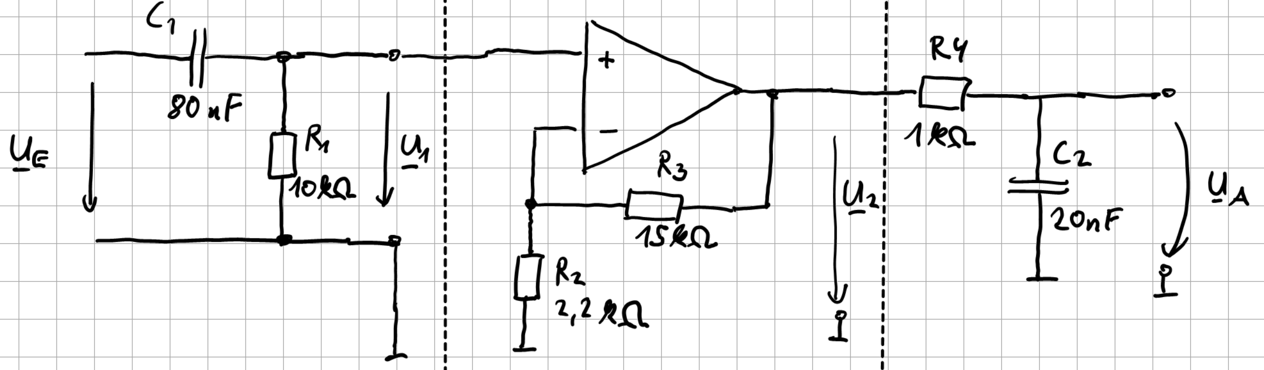
BOD 3



$$\underline{A}_1 = \frac{\underline{U}_1}{\underline{U}_E} = \frac{R_1}{R_1 + \underline{Z}_{C1}}$$

$$= \frac{R_1}{\frac{1}{j\omega C_1} + R_1} \cdot \frac{j\omega C_1}{j\omega C_1}$$

$$= \frac{j\omega \underbrace{C_1 R_1}_{\tau_1}}{1 + j\omega \underbrace{R_1 C_1}_{\tau_1}}$$

$$\tau_1 = \frac{1}{\omega_1}$$

$$\omega_1 = 2\pi f_1$$

$$\underline{A}_1 = \frac{j\frac{\omega}{\omega_1}}{1 + j\frac{\omega}{\omega_1}}$$

$$= \frac{j\,\frac{\cancel{2\pi} f}{\cancel{2\pi} f_1}}{1 + j\,\frac{\cancel{2\pi} f}{\cancel{2\pi} f_1}}$$

Hochpass

$$\boxed{\underline{A}_1 = \frac{j\frac{f}{f_1}}{1 + j\frac{f}{f_1}}}$$

$$\tau_1 = R_1 \cdot C_1$$

$$= 10k\Omega \cdot 80\,nF$$

$$= 800\,\mu s$$

$$\omega_1 = \frac{1}{\tau_1} = 1250\ \frac{1}{s}$$

$$\underline{\underline{f_1}} = \frac{\omega_1}{2\pi} = \underline{\underline{199\ Hz}}$$

$$\underline{A}_2 = \frac{\underline{U}_2}{\underline{U}_1} = 1 + \frac{R_3}{R_2}$$

$$= 1 + \frac{15k\Omega}{2{,}2k\Omega}$$

$$= 7{,}82$$

Konstante

$$K_{dB} = 20 \cdot \log 7{,}82$$

$$= \underline{\underline{17{,}9\ dB}}$$

Aufgabe: